

Satya Nandivada

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EDUCATION

Northeastern University

Master's in Electrical and Computer engineering

January 2023 - May 2025

Gitam University,

Bachelors in Electronics and Communication Engineering

March 2017 – April 2021

EXPERIENCE

DeepOMAP L.L.C, Portland, Maine

Founding Engineer | Computational Biologist (Volunteer)

March 2025 - Present

- Designed **multi-omics AI pipelines** integrating **10x Genomics scRNA-seq, CITE-seq, and ATAC-seq** data for AML subtype prediction, achieving $R^2 = 0.75$ and **0.89 mean Pearson correlation** across 279 ADT markers.
- Implemented **Graph Attention (GAT)** and **Transformer fusion architectures** with dynamic device allocation and GPU cache optimization, delivering **2× faster training** and **35% lower GPU memory use**.
- Orchestrated **Nextflow-based workflows** for data ingestion, QC, and feature extraction; containerized with **Docker/Singularity** and deployed on **AWS and GCP** using CI/CD for reproducibility.
- Conducted **multi-omic integration and network inference** (RNA-protein-chromatin coupling, eQTL & caQTL mapping) to identify regulatory circuits underlying AML heterogeneity.
- Developed **automated diagnostic tools** for pipeline health monitoring, reducing runtime failures by **>60%** and streamlining debug time to minutes

Vanaja Systems Biology lab, Northeastern University

Computational Biology Research Assistant/ Fall-Coop (2024)

March 2023 – Present

- Processed **10x Genomics RNA-seq and ATAC-seq** datasets using **STAR, Kallisto, FastQC, MultiQC, MACS2, DiffBind, and DESeq2**, producing high-quality transcriptomic and epigenomic matrices for downstream modeling
- Built end-to-end **Nextflow/WDL pipelines** with integrated QC (Picard, RSeQC), variant calling (GATK, bcftools, DeepVariant), and differential analysis, ensuring full **FAIR data lineage and provenance**
- Developed **ODE-based mechanistic models** of the **MAPK/PI3K** signaling network in MATLAB for drug-resistance prediction; performed parameter optimization with **>90%** fitting accuracy
- Presented technical insights to cross-disciplinary teams, translating complex computational methods for diverse scientific audiences

Gamer Alliance Members of Esports (GAME LLC) – Remote

Full-Stack AI Engineer

Jul 2025 – Present

- Architected scalable **NestJS + PostgreSQL** infrastructure with modular design, connection pooling, and structured logging for millions of records.
- Architected and deployed a scalable **AWS EC2** instance to host a customized chatbot feature, designing a **FLASK**-based REST API that handled 2,000+ concurrent user sessions with **<200 ms** response latency, improving backend–frontend data flow efficiency by 30% through optimized request routing and caching
- Integrated **AI-based onboarding and data analytics pipelines**, improving request throughput by **4×** and latency by **>45%**.
- Automated performance diagnostics across distributed microservices, reducing critical downtime incidents by **70%**.

Conduent Inc, Hyderabad, India

Application Developer-1, Full-Time

July 2021 – December 2022

- Developed and maintained enterprise **React.js** and **TypeScript** front-end solutions, integrated with **Spring Boot** and Java backends, ensuring sub-second response times for over **10,000** concurrent Medicaid Suite users.
- Designed and launched scalable **GraphQL** and **REST APIs**, powering client portals that reduced average data retrieval latency by 40% and supported **>99.99%** uptime in production environments
- Leveraged **Apache Spark** and **Hadoop** containerized with **Kubernetes** to boost analytics throughput by **3.5×** and accelerate pipelines up to **37×**, while enabling zero-downtime deployments and improved fault tolerance across three data centers.

CONFERENCES

- Poster and Flash Talk Presentation:** A complete mathematical model of the MAPK pathway predicts mechanisms of resistance to BRAF inhibition in BRAF V600E-driven tumors. International Conference of Systems Biology (**ICSB 2024**), IIT Mumbai, India. December 2024.